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Last-mile delivery: Optimal locker location under multinomial logit choice model

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ABSTRACT

One innovative solution to the last-mile delivery problem is the self-service locker system. Motivated by a real case in Singapore, we consider a POP-Locker Alliance who operates a set of POP-stations and wishes to improve the last-mile delivery by opening new locker facilities. We propose a quantitative approach to determine the optimal locker location with the objective to maximize the overall service provided by the alliance. Customer's choices regarding the use of facilities are explicitly considered. They are predicted by a multinomial logit model. We then formulate the location problem as a multi-ratio linear-fractional 0–1 program and provide two solution approaches. The first one is to reformulate the original problem as a mixed-integer linear program, which is further strengthened using conditional McCormick inequalities. This approach is an exact method, developed for small-scale problems. For large-scale problems, we propose an alternating algorithm, i.e., Quadratic Transform with Linear Alternating (QT-LA). The numerical experiment indicates that QT-LA is an efficient approach that yields high-quality solutions. Finally, we conducted a case study. The results highlighted the importance of considering the customers' choices. Under different parameter values of the multinomial logit model, the decisions could be completely different. Therefore, the parameter value should be carefully estimated in advance.

1. Introduction

In recent years, the e-commerce market has been expanding at a remarkable rate. E-commerce revenues worldwide are projected to grow to 6.54 trillion US dollars in 2022, according to Statista, a global business data platform.[1] The rapid growth of e-commerce has led to a sharp increase in the delivery of online purchasing parcels. The process of transporting goods to customers consists of many stages that have undergone significant improvements towards a more efficient and cost-effective manner. However, the final stage, i.e., delivering the parcels (goods) to the doorstep of the customer, is still *not* efficient. In practice, the last-mile delivery (LMD) often involves one package per door (which disallows the economies of scale) and the couriers may have difficulties in finding the exact home address of the related customer. Meanwhile, the problem of "Not-at-Home" can result in an empty trip, especially when the delivery needs a signature of a receipt to confirm the success of a delivery (Deutsch and Golany, 2018). Kedia et al. (2017) showed that missing a delivery may also require a pick-up from a collection point with unsuitable location or time requirements. These issues

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[1] https://www.statista.com/statistics/379046/worldwide-retail-e-commerce-sales/.

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Fig. 1. POP-station at Ang Mo Kio Central, Singapore.

lower the efficiency of the LMD and make the delivery process costly. Therefore, how to address these issues is an essential topic for e-commerce delivery. As a response, operators have started to offer alternative options to regular home deliveries.

Self-service technology (SST) provides an innovative solution. In recent years, self-service POP-stations and parcel lockers (see Fig. 1) have been deployed for the LMD. They allow logistics companies to deliver parcels to the facilities with economies of scale. Customers can collect the parcels at their convenience. Recent research showed that self-service POP-stations and parcel locker networks are gaining more popularities (Dieke et al., 2013; Morganti et al., 2014; Wang et al., 2016) and their deployments significantly improve the consumers' shopping experience and enhance competitiveness and performance of the business (Vakulenko et al., 2018; Wang et al., 2020). In Singapore, Singapore Post has launched the "Locker Alliance" program. The alliance provides an extensive delivery network that consists of self-service POP-stations and parcel lockers. Using this system, logistics companies can avoid the "Not-at-Home" problem in deliveries and customers do not need to wait at home for receiving the parcels. Therefore, it creates a "win-win" situation for companies and customers.

There is a proliferation of research on the parcel locker. Many studies have highlighted the benefits of using it from both economic and environmental aspects. For instance, Kämäräinen et al. (2001) used delivery data from the suburban area of Helsinki and compared the delivery costs of two strategies, namely, the regular home delivery and the batch delivery to reception boxes. Their results showed a 42% cost reduction when the deliveries are made to boxes. Similarly, Punakivi et al. (2001) conducted a case study on a retail company in Finland and showed that the operational cost of regular home delivery is more expensive than that of delivery to reception boxes and parcel lockers. Song et al. (2013) and Lemke et al. (2016) argued that the greenhouse gas emissions can be reduced by deploying the self-collection service (e.g., parcel lockers or collection-and-delivery points). Researchers have also examined the factors influencing customers' intention to use self-collection services. Xu and Hong (2013) showed that implementing customer segmentation in self-collection services can better suit customers' need and customers' satisfaction of home delivery service is not a significant factor that influences customers' willingness to use self-collection services. Using survey data in Singapore, Yuen et al. (2018) argued that, to improve customers' intention, it is important is to integrate self-collection services into customers' lifestyle, values and needs. Other studies focused on the conceptualization of hyperconnected lockers network (Faugere and Montreuil, 2017a) and configuration designs of locker banks based on optimization models (Faugere and Montreuil, 2017b; Faugère and Montreuil, 2020). Among the literature, Deutsch and Golany (2018) are the first to propose a quantitative approach to determine the optimal locker location and the sizes of lockers. They assumed that customers made choices in a deterministic fashion and formulated an optimization problem to maximize the total profit of the operator. More recently, Schwerdfeger and Boysen (2020) studied the dynamic location model of mobile parcel lockers. They assumed that lockers can change their locations during the day and optimized the locations of lockers so that they are accessible to customers within a predefined range. The objective is to minimize the number of mobile lockers while ensuring that all customers are served.

In this paper, we study the locker location problem. Motivated by the "Locker Alliance" program in Singapore, we consider an alliance scenario where two operators corporate with each other to maximize the overall service level. We assume that the demand for the locker service is fixed and the service level for a customer who uses a facility is a decreasing function of the walking distance. The problem is not only to decide where to locate the new lockers, but also to close some unnecessary facilities whose deployments can reduce the overall service level when new lockers come to the network. Different from Deutsch and Golany (2018) and Schwerdfeger and Boysen (2020), we use a random utility model to predict customer's choices. In practice, customers may prefer some facility (not necessarily the most convenient one) according to their preferences. The preferences are, however, unknown to the operators (Ljubić and Moreno, 2018). Empirical and experimental studies also show that individual choice behaviors are difficult to capture and they are typically interpreted probabilistically (Gul et al., 2014). One way to predict the customer's choices is to apply the random utility theory. More specifically, we use the multinomial logit (MNL) model because we have witnessed its successful applications in various facility location problems where the customer's choices are considered. Aros-Vera et al. (2013) addressed the

park-and-ride facility location problem using the MNL framework. In the healthcare facility location problem, Zhang et al. (2012) considered a probabilistic-choice model based on the MNL, which states a client may patronize each facility with a certain probability that increases with the attractiveness. In a subsequent model, the probabilistic choice was assumed to be determined by the travel time to a facility and the expected waiting time at a facility (Zhang and Atkins, 2019). In the maximum capture problem with random utilities, the MNL was used to forecast the capture demand. The problem considers a company entering a market and seeking to locate facilities such that the captured demand is maximized. Recent works on this area focus on developing solution approaches. For example, Haase and Müller (2014) discussed different linear reformulation techniques, while Freire et al. (2016) studied various specialized branch-and-bound algorithms. Ljubić and Moreno (2018) proposed branch-and-cut approaches based on outer approximation and submodular cuts. Later, a multi-cut outer approximation algorithm was developed (Mai and Lodi, 2020).

Traditionally, the location problem with customer choices was modeled as a mixed-integer convex program (Haase and Müller, 2014; Ljubić and Moreno, 2018; Lin and Tian, 2020; Mai and Lodi, 2020). However, in order to reflect the reduction in service level when the walking distance increases and model the possibility of closing some existing facilities, our problem becomes more complicated and is formulated as a mixed-integer non-convex program, which is indeed a multi-ratio linear-fractional 0–1 program (MLFP) (Borrero et al., 2017). To solve a MLFP, the most straightforward approach is to reformulate it as an equivalent mixed-integer linear programming (MILP) (Borrero et al., 2017). The major drawback of this approach is that it cannot scale well on problem size and may fail to solve a large-scale (even just a moderate-scale) problem in a reasonable time. On the other hand, one exact approach that is capable of solving large-scale MLFPs is the outer approximation algorithm (OA) (Elhedhli, 2005; Mai and Lodi, 2020), which proceeds by solving a sequence of MILPs with an increasing number of constraints that refine the feasible region. The OA algorithm guarantees finite convergence and outperforms the MILP approach by a large margin in general. However, it requires the concavity of the objective function, which makes it not valid for our problem because our objective function is not concave. Therefore, the quadratic transform approach, which transforms the MLFP into a bi-concave maximization program (BCMP), has been developed to solve such issue (Benson, 2004; Shen and Yu, 2018). But solving the BCMP itself requires significant efforts to design a specific algorithm (Benson, 2007) and the solving process can be time-consuming as well. For comprehensive survey papers on the applications and algorithms for MLFPs, we refer the reader to Schaible and Shi (2003) and Borrero et al. (2017). In this paper, we apply the quadratic transform approach and design an efficient alternating algorithm to solve large-scale BCMPs.

To summarize, our contributions are fourfold. The first one is that we consider a POP-Locker Alliance who operates a set of POP-stations and wishes to improve the service level by opening new locker facilities and closing unnecessary (if any) POP-stations. We predict customer's choices by a MNL model and propose a MLFP to determine the optimal locker location. Even though the MNL has been applied in previous facility location problems (Haase and Müller, 2014; Ljubić and Moreno, 2018; Mai and Lodi, 2020), this paper is the first to consider the probabilistic choice in the locker network design and our model is a non-concave maximization model, which is inherently more challenging. Our second contribution is to reformulate the original problem as an equivalent MILP, which is further strengthened using McCormick inequalities. This approach is an exact method that works well for small-scale problems. However, it cannot scale well on the problem size. The poor performance of MILPs on large-scale problems motivates our third contribution, i.e., we propose the *Quadratic Transform with Linear Alternating* (QT-LA) algorithm to efficiently find high-quality solutions. Numerical studies indicated that our proposed algorithm works well. Finally, we conduct a case study based on the POP-locker system in Singapore and provide managerial implications.

The rest of the paper is organized as follows. Section 2 covers the problem description and the basic mathematical model. We present the MILP solution method in Section 3 and QT-LA in Section 4. Numerical studies are conducted in Section 5. We conclude the paper in Section 6.

Table 1
Nomenclature.

Sets		
I	:	set of customer zones
J	:	set of candidate locker facilities
K	:	set of POP-stations
Parameters		
l_{ik}	:	distance between demand zone i and pos-station k, $\forall\, i \in I, k \in K$
L_{ij}	:	distance between demand zone i and locker j, $\forall\, i \in I, j \in J$
$\tilde{a}_{ik}$	:	service level when a customer in zone i uses POP-station k. $\tilde{a}_{ik} \in [0, 1]$, $\forall\, i \in I, k \in K$
a_{ij}	:	service level when a customer in zone i uses locker j. $a_{ij} \in [0, 1]$, $\forall\, i \in I, j \in J$
v_{im}	:	utility when a customer in zone i uses facility m, $\forall\, m \in J \cup K$
$\tilde{\theta}_{ik}$	:	preference of a customer in zone i to POP-station k, $\forall\, k \in K$
θ_{ij}	:	preference of a customer in zone i to locker j, $\forall\, j \in J$
d_i	:	proportion of the total demand that arises from zone i, $\forall\, i \in I$
P	:	maximum number of open lockers/ maximum number of closed POP-stations
Decision Variables		
x_j	:	1, if locker j is open; 0, otherwise. $\forall\, i \in I$
r_k	:	1, if POP-station k remains open; 0, otherwise. $\forall\, k \in K$

2. Problem description

In this section, we present the problem description. Table 1 gives the notations. Additional notations will be introduced when necessary.

Consider a POP-Locker Alliance that operates a set of POP-stations (denoted by set K) to serve a set of customer zones (denoted by set I). The customers are assumed homogeneous in their observable characteristics. The distance between customer zone i and POP-station k is l_{ik}, $\forall\, i \in I$, $k \in K$. The service level that a customer in zone i patronizes POP-station k is measured by $\widetilde{a}_{ik} \in [0, 1]$. We assume $\widetilde{a}_{ik}$ is a nondecreasing function of l_{ik}, i.e., when the distance between node i and node k increases, the service level should not increase. Besides POP-stations, there are potential locker facilities (denoted by set J). The distance between customer zone i and locker j is L_{ij}, $\forall\, i \in I$, $j \in J$. The service level that a customer in zone i patronizes locker j is measured by $a_{ij} \in [0, 1]$. Similarly, a_{ij} is a nondecreasing function of L_{ij}.

The alliance wishes to open new locker facilities and to close some unnecessary (if any) POP-stations so that the overall service level provided by the POP-Locker system is maximized. If the alliance assigns customers to facilities or customers always select the nearest facility, the problem becomes a type of classical coverage problems (Daskin, 2011). It is then advantageous to open P locker facilities and not to close any POP-station when we restrict the maximum number of open locker facilities to P. However, customers are taking the initiative in e-commence. They may prefer a facility that is not the nearest according to their preferences. The preferences are, however, unknown and hard to observe (Ljubić and Moreno, 2018). Empirical and experimental evidences have supported that individual choice behaviors should be interpreted probabilistically (Gul et al., 2014). In recent studies on the parcel locker and facility location problems, the probabilistic choice models have been validated using big-data (Lyu and Teo, 2019; Suhara et al., 2019; Gunawan et al., 2020). Therefore, the results from the traditional coverage model may not be relevant and the probabilistic choice behaviors should be considered.

The multinomial logit model (MNL) (McFadden, 1973) provides a way to forecast the discrete choice behaviors. It assumes that customers in zone i will maximize their utility:

$$u_{im} = v_{im} + \epsilon_{im},\ \forall\, i \in I,\ m \in J \cup K \tag{1}$$

when choosing a facility. Here, v_{im} is the deterministic part of the utility, which is measured as a function of convenience (distance). ϵ_{im} is a random term that is independent identically extreme value distributed.

Define a binary variable r_k, $\forall\, k \in K$, such that $r_k = 1$, if POP-station k remains in operation; $r_k = 0$, otherwise. Define a binary variable x_j, $\forall\, j \in J$, such that $x_j = 1$, if locker j is open; $x_j = 0$, otherwise. According to MNL, the probability that a customer in zone i selects POP-station k is

$$p_{ik} = \frac{e^{v_{ik}} r_k}{\sum_{k\in K} e^{v_{ik}} r_k + \sum_{j\in J} e^{v_{ij}} x_j} \tag{2}$$

Similarly, the probability that a customer in zone i selects locker j is

$$p_{ij} = \frac{e^{v_{ij}} x_j}{\sum_{k\in K} e^{v_{ik}} r_k + \sum_{j\in J} e^{v_{ij}} x_j} \tag{3}$$

Let $\widetilde{\theta}_{ik} = e^{v_{ik}}$, $\forall\, k \in K$ and $\theta_{ij} = e^{v_{ij}}$, $\forall\, i \in I$. The proportion of total customers arising from zone i and using POP-station k is:

$$D_{ik} = d_j p_{ik} = \frac{d_i \widetilde{\theta}_{ik} r_k}{\sum_{k\in K} \widetilde{\theta}_{ik} r_k + \sum_{j\in J} \theta_{ij} x_j} \tag{4}$$

Similarly, the proportion of total customers arising from zone i and using locker j is:

$$D_{ij} = d_j p_{ij} = \frac{d_i \theta_{ij} x_j}{\sum_{k\in K} \widetilde{\theta}_{ik} r_k + \sum_{j\in J} \theta_{ij} x_j} \tag{5}$$

We then compute the overall service level by

$$C\left(x, r\right) = \sum_{i\in I}\left(\sum_{k\in K} \widetilde{a}_{ik} D_{ik} + \sum_{j\in J} a_{ij} D_{ij}\right) = \sum_{i\in I} \frac{d_i\left(\sum_{k\in K} \widetilde{a}_{ik}\widetilde{\theta}_{ik} r_k + \sum_{j\in J} a_{ij}\theta_{ij} x_j\right)}{\sum_{k\in K} \widetilde{\theta}_{ik} r_k + \sum_{j\in J} \theta_{ij} x_j} \tag{6}$$

It is easy to see that $C(x, r)$ lies between 0 and 1. To facilitate later discussions, we define parameters $\widetilde{b}_{ik}$ and b_{ij} such that $\widetilde{b}_{ik} = d_i \widetilde{a}_{ik} \widetilde{\theta}_{ik}$ and $b_{ij} = d_i a_{ij} \theta_{ij}$. With them, we propose the following model for determining the optimal locker location under MNL model:

$$\left[\mathbf{P1}\right] \quad \max C\left(x, r\right) = \sum_{i\in I} \frac{\sum_{k\in K} \widetilde{b}_{ik} r_k + \sum_{j\in J} b_{ij} x_j}{\sum_{k\in K} \widetilde{\theta}_{ik} r_k + \sum_{j\in J} \theta_{ij} x_j} \tag{7}$$

$$st. \sum_{j \in J} x_j \leqslant P \tag{8}$$

$$\sum_{k \in K} \left(1 - r_k\right) \leqslant P \tag{9}$$

$$x_j \in \{0, 1\}, \forall\, j \in J \tag{10}$$

$$r_k \in \{0, 1\}, \forall\, k \in K \tag{11}$$

where $0 \leqslant P \leqslant \max\{|J|, |K|\}$. The objective is to maximize the overall service level $C(x, r)$. Constraint (8) imposes that the maximum number of open lockers is P. Constraint (9) indicates that the maximum number of closed POP-stations cannot exceed P. The maximum number of open lockers and the maximum number of closed POP-stations need not be the same. For simplicity and without loss of generality, we assume a same value for both constraints. Meanwhile, if constraints (8) and (9) are set to equalities, then they simply indicate P POP-stations will be replaced by the same number of lockers in the network. Finally, constraints (10) and (11) specify that x and r are binary variables.

Remark 1. In [P1], the utility and the service level are measured by distance. The MNL is developed based on this assumption. In practice, other dimensions are worth investigating as well. Using empirical studies, Yuen et al. (2019) showed that customers' intention to use lockers are affected by factors such as convenience, privacy security, and service reliability. Lyu and Teo (2019) suggested that locker types are also statistically significant to influence customer choices. Therefore, incorporating these dimensions in the formulation will make [P1] more relevant to practice. Nevertheless, [P1] provides a model framework that is ready to be generalized to more complicated utility functions, given that the impacts of other dimensions on utility are available, or they can be calibrated from real-world datasets.

Remark 2. One important assumption for [P1] is that the demand for the locker service is fixed. Therefore, the competition effect between locker service and other delivery modes is not considered. The goal is to maximize the service level. This perspective is fundamentally different from other facility location problems employing probabilistic choice rules, in which the demand attracted to facilities is a variable and the objective is to win the competition against the other options (Haase and Müller, 2014; Ljubić and Moreno, 2018; Mai and Lodi, 2020).

3. Mixed-integer linear programming approach

In this section, we present an exact solution approach that works well for small-scale problems. We first derive an equivalent MILP formulation for P1 and then strengthen the formulation by conditional McCormick inequalities.

3.1. Basic formulation

The MILP reformulation approach is widely used for multi-ratio linear fractional 0–1 programing (Bront et al., 2009; Haase and Müller, 2014; Méndez-Díaz et al., 2014). Given recent improvements of off-the-shelf MILP solvers such as CPLEX and Gurobi, reformulating P1 as its equivalent MILP provides a standard solution framework that is straightforward to implement.

Let $z_i = 1/\left(\sum_{k \in K} \widetilde{\theta}_{ik} r_k + \sum_{j \in J} \theta_{ij} x_j\right)$. P1 is equivalent to the bilinear mixed 0–1 problem:

$$\max \sum_{i \in I} \sum_{k \in K} \widetilde{b}_{ik} r_k z_i + \sum_{i \in I} \sum_{j \in J} b_{ij} x_j z_i \tag{12}$$

$$st. \sum_{k \in K} \widetilde{\theta}_{ik} r_k z_i + \sum_{j \in J} \theta_{ij} x_j z_i = 1, \forall\, i \in I \tag{13}$$

$$z_i \geqslant 0, \forall\, i \in I \tag{14}$$

(8)–(11)

which can be recast as a MILP by linearizing the bilinear terms $x_j z_i$ and $r_k z_i$. Both terms involve the product of a binary variable and a continuous, nonnegative variable. They can be transformed into (mixed-integer) linear constraints (Chen et al., 2010). Let $y_{ij} = x_j z_i$ and $Y_{ik} = r_k z_i$. Consider the following MILP:

$$[\textbf{MILP}] \quad \max \sum_{i \in I} \sum_{k \in K} \widetilde{b}_{ik} Y_{ik} + \sum_{i \in I} \sum_{j \in J} b_{ij} y_{ij} \tag{15}$$

$$st. \sum_{k \in K} \widetilde{\theta}_{ik} Y_{ik} + \sum_{j \in J} \theta_{ij} y_{ij} = 1, \forall\, i \in I \tag{16}$$

$$z_i - U(1 - x_j) \leqslant y_{ij} \leqslant z_i, \forall\, i \in I, j \in J \tag{17}$$

$$0 \leqslant y_{ij} \leqslant Ux_j, \forall\, i \in I, j \in J \tag{18}$$

$$z_i - U(1 - r_k) \leqslant Y_{ik} \leqslant z_i, \forall\, i \in I, k \in K \tag{19}$$

$$0 \leqslant Y_{ik} \leqslant Ur_k, \forall\, i \in I, k \in K \tag{20}$$

(8)–(11), (14)

where U be a sufficiently large number. It can easily verified that this MILP is equivalent to P1. Specifically, (17), (18) impose that $y_{ij} = x_j z_i$. When $x_j = 1$, $z_i = y_{ij}$ by (17); when $x_j = 0$, $z_i = 0$ by (18). Similarly, (19), (20) impose that $Y_{ik} = r_k z_i$.

This approach is straightforward. In theory, the MILP can yield the same optimal solution as P1. However, as shown in Bront et al. (2009) and Méndez-Díaz et al. (2014), the MILP does not scale well on the problem size. In practice, it may fail to yield the optimal solution in a reasonable computational time, even for moderate-size problems.

3.2. Strengthened formulation: conditional McCormick inequalities

To enhance the performance of the MILP, we strengthen the MILP formulation using (conditional) McCormick estimators for the bilinear terms. The fundamental idea is to tighten the relaxation bound. We look at the following proposition.

Proposition 1. *A bilinear mixed 0–1 term $y = xz$, where x is a 0–1 variable and z is a continuous variable with $z > 0$, can be exactly reformulated as the following linear inequalities: (a) $z - z^u \cdot (1 - x) \leqslant y \leqslant z - z^l|_{x=0} \cdot (1 - x)$; (b) $z^l|_{x=1} \cdot x \leqslant y \leqslant z^u|_{x=1} \cdot x$, where z^u is the global upper bound on z;$z^l|_{x=0}$ is the lower bound on z when $x = 0$;$z^l|_{x=1}$ and $z^u|_{x=1}$ are the lower bound and upper bound on z when $x = 1$.*

Proof. We need to check two cases:

- If $x = 0$, we have: (a) $z - z^u \leqslant y \leqslant z - z^l|_{x=0}$; (b) $0 \leqslant y \leqslant 0$. Clearly, (a) is inactive because $-z^u \leqslant -z^l|_{x=0}$ and (b) enforces that $y = 0$.
- If $x = 1$, we have: (a) $z \leqslant y \leqslant z$; (b) $z^l|_{x=1} \leqslant y \leqslant z^u|_{x=1}$. Clearly, (b) is inactive because $z^l|_{x=1} \leqslant z^u|_{x=1}$ and (a) enforces that $y = z$.

Combine the cases, inequalities (a) and (b) give $y = xz$. □

By Proposition 1, the bilinear terms $x_j z_i$ and $r_k z_i$ with x_j and r_k being binary variables can be represented by linear inequalities with global and conditional bounds on z_i subject to (8)–(11). We demonstrate that bounds on z_i can be obtained analytically.

We first give the global upper bound on z_i.

Proposition 2. *The following global upper bounds on z_i, $\forall\, i \in I$, is valid:*

$$z_i^u = \frac{1}{\sum_{e=1}^{|K|-P} \widetilde{\theta}_{i(e)}} \tag{21}$$

where $\widetilde{\theta}_{i(e)}$ is the eth smallest value of $\widetilde{\theta}_{in}$, $\forall\, n \in K$.

Proof. See Appendix A. □

Next, we derive the conditional bounds. Let $z^u_{i|xj=\xi}$ and $z^l_{i|xj=\xi}$ be the upper and lower bound on z_i when an additional constraint $x_j = \xi$ is imposed.

Proposition 3. *The following conditional bounds on z_i given x_j , $\forall\, i \in I, j \in J$, are valid:*

$$z^u_{i|xj=1} = \frac{1}{\sum_{e=1}^{|K|-P} \widetilde{\theta}_{i(e)} + \theta_{ij}} \tag{22}$$

$$z^l_{i|xj=0} = \frac{1}{\sum_{k\in K} \widetilde{\theta}_{ik} + \sum_{e=1}^{P} \theta_{i[e]}} \tag{23}$$

$$z^l_{i|xj=1} = \frac{1}{\sum_{k\in K} \widetilde{\theta}_{ik} + \theta_{ij} + \sum_{e=1}^{P-1} \theta_{i[e]}} \tag{24}$$

where $\widetilde{\theta}_{i(e)}$ is the eth smallest value of $\widetilde{\theta}_{in}$, $\forall\, n \in K$;$\theta_{i[e]}$ is the eth largest value of θ_{im}, $\forall\, m \in J \setminus \{j\}$.

Proof. See Appendix A. □

Similarly, let $z^u_{i|r_k=\xi}$ and $z^l_{i|r_k=\xi}$ be the upper and lower bound on z_i when an additional constraint $r_k = \xi$ is imposed.

Proposition 4. *The following conditional bounds on z_i given r_k , $\forall\, i \in I$, $k \in K$, are valid:*

$$z^u_{i|r_k=1} = \frac{1}{\sum_{e=1}^{|K|-P-1} \widetilde{\theta}_{i(e)} + \widetilde{\theta}_{ik}} \tag{25}$$

$$z^l_{i|r_k=0} = \frac{1}{\sum_{k\in K\setminus\{k\}} \widetilde{\theta}_{ik} + \sum_{e=1}^{P} \theta_{i[e]}} \tag{26}$$

$$z^l_{i|r_k=1} = \frac{1}{\sum_{k\in K} \widetilde{\theta}_{ik} + \sum_{e=1}^{P} \theta_{i[e]}} \tag{27}$$

where $\widetilde{\theta}_{i(e)}$ is the eth smallest value of $\widetilde{\theta}_{in}$, $\forall\, n \in K\setminus\{k\}$;$\theta_{i[e]}$ is the eth largest value of θ_{im}, $\forall\, m \in J$.

Proof. See Appendix A. □

With the above bounds on z_i, we can define the following (conditional) McCormick inequalities (McCormick, 1976) for $y_{ij} = x_j z_i$ and $Y_{ik} = r_k z_i$:

$$z_i - z_i^u \cdot \left(1 - x_j\right) \leqslant y_{ij} \leqslant z_i - z^l_{i|xj=0} \cdot \left(1 - x_j\right),\ \forall\, i \in I, j \in J \tag{28}$$

$$[MC] \quad z^l_{i|xj=1} \cdot x_j \leqslant y_{ij} \leqslant z^u_{i|xj=1} \cdot x_j,\ \forall\, i \in I, j \in J \tag{29}$$

$$z_i - z_i^u \cdot (1 - r_k) \leqslant Y_{ij} \leqslant z_i - z^l_{i|r_k=0} \cdot (1 - r_k),\ \forall\, i \in I, k \in K \tag{30}$$

$$z^l_{i|r_k=1} \cdot r_k \leqslant Y_{ij} \leqslant z^u_{i|r_k=1} \cdot r_k,\ \forall\, i \in I, k \in K \tag{31}$$

which is an exact reformulation. One can observe that the polyhedron defined by the continuous relaxation of (28)–(31) is a subset of that by (17)–(20). Adding the MC can tighten the relaxation bound, thereby avoiding extensive branching during the branch-and-bound procedure and speeding up the computation in general (Şen et al., 2018). The impacts will be discussed in Section 5.1.1.

4. Quadratic transform with linear alternating

In Section 3, we present the MILP approach to P1. It is a favored approach for small-scale problems because it is an exact method that is easy to implement. However, the MILP has significantly more variables and constraints than P1. The solution times for large-scale problems can be prohibitive. To solve large-scale problems, we propose an algorithm that efficiently generates high-quality solutions. We start with the following proposition:

Proposition 5. *P1 is equivalent to the following problem:*

$$\left[\mathbf{P2}\right] \quad \max \sum_{i\in I} 2y_i \sqrt{\sum_{k\in K} \widetilde{b}_{ik} r_k + \sum_{j\in J} b_{ij} x_j} - \sum_{i\in I} y_i^2 \left(\sum_{k\in K} \widetilde{\theta}_{ik} r_k + \sum_{j\in J} \theta_{ij} x_j \right) \tag{32}$$

$$st.\ y_i \geqslant 0,\ \forall\, i \in I \tag{33}$$

$$(8)\text{–}(11)$$

Proof. See Benson (2004). □

In effect, the equivalence of P1 and P2 can be easily established: we first optimize P2 over y_i. Clearly, this partial optimization is a concave maximization problem. By the first-order condition, we have:

$$y_i := \frac{\sqrt{\sum_{k\in K} \widetilde{b}_{ik} r_k + \sum_{j\in J} b_{ij} x_j}}{\sum_{k\in K} \widetilde{\theta}_{ik} r_k + \sum_{j\in J} \theta_{ij} x_j},\ \forall\, i \in I \tag{34}$$

Plugging (34) into (32) leads us to (7).

According to Shen and Yu (2018), P2 is also referred to as the Quadratic Transform (QT). Obviously, P2 is not a convex optimization problem. It is indeed a BCMP: when x and r are held fixed, the optimal y can be found in closed form as (34). When y is held fixed, P2 reduces to a (nonlinear) concave maximization program with variables x and r.

One heuristic solution approach that leverages the bi-concavity of QT is to alternate between updating y and solving the nonlinear program until some predetermined stopping condition is met, as proposed by Shen and Yu (2018). However, this approach is initially

proposed for continuous problems. When it is applied to 0–1 problems, it may stop at low-quality solutions or even loop around candidate solutions without convergence. One reason for its poor performance is the "aggressive" rule for updating y: y only depends on the current solution of x and r by (34). Meanwhile, in our problem, solving the nonlinear 0–1 program for multiple iterations can be expensive when the problem size is large.

To overcome the issues, we propose a new alternating algorithm. Define function $f_i(x, r)$ such that

$$f_i\left(x, r\right) = \sqrt{\sum_{k\in K} \widetilde{b}_{ik} r_k + \sum_{j\in J} b_{ij} x_j} \tag{35}$$

We can rewrite P2 in the hypograph form of f_i, i.e.,

$$\left[\mathbf{P3}\right] \quad \max \sum_{i\in I} 2y_i\beta_i - \sum_{i\in I} y_i^2\left(\sum_{k\in K} \widetilde{\theta}_{ik} r_k + \sum_{j\in J} \theta_{ij} x_j\right) \tag{36}$$

$$st.\ \beta_i \leqslant f_i(x, r) \tag{37}$$

(8)–(11)

Now, given any point $(\bar{x}, \bar{r})$, since $f_i(x, r)$ is a concave function, we can bound it from above by its first-order linear approximation on $(\bar{x}, \bar{r})$. The following constraint is thus valid for P3:

$$\beta_i \leqslant \sum_{j\in J} \frac{\partial f_i(\bar{x}, \bar{r})}{\partial x_j}\left(x_j - \bar{x}_j\right) + \sum_{k\in K} \frac{\partial f_i(\bar{x}, \bar{r})}{\partial r_k}\left(r_k - \bar{r}_k\right) + f_i\left(\bar{x}, \bar{r}\right), \forall\ i \in I \tag{38}$$

where $\frac{\partial f_i(\bar{x}, \bar{r})}{\partial x_j}$ and $\frac{\partial f_i(\bar{x}, \bar{r})}{\partial r_k}$ are the partial derivative of f_i with respect to x_j and r_k evaluated at $(\bar{x}, \bar{r})$. Clearly, any feasible point satisfying (37) is also feasible in the region defined by (38), i.e., (38) does not eliminate any feasible region of P3. We can then model P3 using the following MILP formulation:

$$\max \sum_{i\in I} 2y_i\beta_i - \sum_{i\in I} y_i^2\left(\sum_{k\in K} \widetilde{\theta}_{ik} r_k + \sum_{j\in J} \theta_{ij} x_j\right) \tag{39}$$

$$st.\ \beta_i \leqslant \sum_{j\in J} \frac{\partial f_i(x^t, r^t)}{\partial x_j}\left(x_j - x_j^t\right) + \sum_{k\in K} \frac{\partial f_i(x^t, r^t)}{\partial r_k}\left(r_j - r_j^t\right) + f\left(x^t, r^t\right), \forall\ i \in I, \left(x^t, r^t\right) \in T \tag{40}$$

(8)–(11)

where T is the set of recorded points (x, r). As the number of points in T increases, the above formulation provides better approximation for P3.

With the above program, we now present our proposed algorithm in Algorithm 1. We refer to it as *Quadratic Transform with Linear Alternating* (QT-LA) algorithm.

In Step 0, we initialize the solution and parameters. The initial solution need not to be feasible. n is the iteration number and N_{max} is the maximum number of iterations.

In Step 1, we update set T and y and compute the approximated partial derivative. Here, γ serves as the step-size. It is a number between 0 and 1, acting like a stabilization factor that controls the intensity of moving y towards the value defined by (34) at current solution (x^n, y^n). In practice, this parameter is important for the algorithm. A brief discussion of its impacts is presented in Appendix B. When computing the partial derivative, we add a small number ϵ to the denominate. This is to enhance the numerical stability and avoid the zero value in the denominate.

In Step 2, we solve a spare MILP. This can be done using advanced MILP solvers. As the algorithm proceeds, new points (tentative solutions during iterations) are added to set T and thus, the MILP contains more constraints and yields better approximation for the original formulation.

Finally, Step 3 is to check whether the stopping condition is met. The algorithm terminates when it finds repeated solutions or the maximum number of iterations exceeds a predetermined number. If the stopping condition is unmet, the process returns to Step 1.

The effectiveness and efficiency of QT-LA algorithm are discussed in Section 5.1.2. We point out that QT-LA is a heuristic algorithm. As in Algorithm 1, we need to provide an initial solution (x^0, r^0). It is well known that, for an alternating heuristic, the initial condition may affect the convergent point (Gorski et al., 2007). To enhance the solution quality, one can give multiple initial solutions, run QT-LA accordingly, and select the best solution from the pool, i.e., using the multi-start version of QT-LA. However, doing so will definitely increase the computational time because QT-LA will be performed multiple times.

Algorithm 1. QT-LA Algorithm

Step 0: Initialize x^0 and r^0;$y^0 = 0$;$n = 0$;$\epsilon = 0.0001$;$T = \varnothing$; choose $\gamma \in (0, 1]$;$N_{max} \in Z_+$.
Step 1: T: $=T \cup \{(x^n, r^n)\}$. $n := n + 1$ Update parameters:

$$y_i^n := \left(1-\gamma\right)y_i^{n-1} + \gamma.\frac{\sqrt{\sum_{k\in K}\tilde{b}_{ik}r_k^{n-1} + \sum_{j\in J} b_{ij}x_j^{n-1}}}{\sum_{k\in K}\tilde{\theta}_{ik}r_k^{n-1} + \sum_{j\in J}\theta_{ij}x_j^{n-1}}, \forall\, i \in I$$

$$\frac{\partial f_i^n}{\partial x_j} := \frac{b_{ij}}{2\sqrt{\sum_{k\in K}\tilde{b}_{ik}r_k^{n-1} + \sum_{j\in J} b_{ij}x_j^{n-1} + \epsilon}}, \forall\, i \in I, j \in J$$

$$\frac{\partial f_i^n}{\partial r_k} := \frac{\tilde{b}_{ik}}{2\sqrt{\sum_{k\in K}\tilde{b}_{ik}r_k^{n-1} + \sum_{j\in J} b_{ij}x_j^{n-1} + \epsilon}}, \forall\, i \in I, j \in J$$

Step 2: Solve the following MILP:

$$\max \sum_{i\in I} 2y_i^n\beta_i - \sum_{i\in I}(y_i^n)^2\left(\sum_{k\in K}\tilde{\theta}_{ik}r_k + \sum_{j\in J}\theta_{ij}x_j\right)$$

$$st.\ \beta_i \leqslant \sum_{j\in J}\frac{\partial f_i^t}{\partial x_j}\left(x_j - x_j^t\right) + \sum_{k\in K}\frac{\partial f_i^t}{\partial r_k}\left(r_j - r_j^t\right) + f\left(x^t, r^t\right), \forall\, i \in I, \left(x^t, r^t\right) \in T \quad (8) - -(11)$$

to obtain the solution (x^n, r^n).
Step 3: If $(x^n, r^n) \in T$ or $n > N_{max}$, stop and output (x^n, r^n). Else, return to Step 1.

5. Numerical study

In this section, we conduct numerical studies on the performance of different solution approaches. The results show that adding MC to MILP can largely reduce the computational time. However, for large-scale problems, MILP + MC fails to solve them in a resealable time, whereas our proposed QT-LA can efficiently handle them in seconds and leads to better solution quality (compared to the solution found by MILP + MC in 3600 s) in general. Finally, we present a case study based on the POP-Locker system in Singapore.

5.1. Algorithm Performance

We use artificially generated data. The Cartesian coordinates of the node set I, J and K are randomly generated by a uniform distribution in the interval [0, 1000]. The demand of each node in I are randomly generated by a uniform distribution in the interval [0, 100]. The deterministic part of utility is defined as $v_{im} = -\alpha L_{im}$, where L_{im} is the euclidian distance between node i and m, $\forall\, i \in I$, $m \in J \cup K$. The unit of L_{im} is 100. For the value of a_{im}, we adopt the following step-wise function: $a_{im} = 1$, if $L_{im} \leqslant 1$;$a_{im} = 0.5$, if $1 < L_{im} \leqslant 2$;$a_{im} = 0.2$, if $2 < L_{im} \leqslant 3$;$a_{im} = 0$, if $L_{im} > 3$.

We solve the MILPs and QT-LA through the Python-embedded modeling language CVXPY (Diamond and Boyd, 2016) and use Gurobi 8.1.1 under default setting as the solver. All computational experiments are done on a 16 GB memory iOS computer with 2.6 GHz Intel Core i7 processor.

5.1.1. Performance of MILP formulations

We investigate the performance of MILP formulations and impact of adding MC. We use three artificial generated networks, namely, the 60-node network, the 80-node network and the 100-node network. The number of customer zones, POP-stations and candidate lockers is 30, 15 and 15 in the 60-node network; 40, 20 and 20 in the 80-node network; and 50, 25 and 25 in the 100-node network. We set the maximal computational time to 3600 s. Table 2 reports the results.

In general, the computational time declines with α increasing and grows with P increasing. That is, for both formulations, when the value of α is small and P is large, the problem instances require significantly more computational efforts than those with large α value and small P value.

In almost all problem instances, adding the MC can lead to faster computation or smaller MIP gap, especially when the instances are difficult to solve by MILP. For example, in the 80-node network, when $\alpha = 0.5$ and $P = 5$, MILP requires 1451.8s to find the global optimal solution, while MILP + MC requires only 114.5s. Meanwhile, when the instances cannot be solved, MILP + MC terminates with much smaller gaps. The better performance of MILP + MC is due to the stronger continuous relaxation. Therefore, when implementing the MILP approach, it is important to add the MC.

Finally, the results suggest both formulations cannot scale well on the problem size. In the 60-node network, all problem instances can be solved easily using both formulations. The maximum computational times among these 18 instances are 150.6s and 55s for

Table 2
Computational time and Gap via MILP and MILP + MC under the 60-node, 80-node and 100-node problem instances. CPU denotes the times in seconds used to solve the respective problem. The maximal computational time is set to 3600 seconds. ** indicates the instance cannot be solved to global optimality. Gap is the MIP gap reported by Gurobi in %.

α	P	60-node				80-node				100-node			
		MILP		MILP + MC		MILP		MILP + MC		MILP		MILP + MC	
		CPU	Gap	CPU	Gap	CPU	Gap	CPU	Gap	CPU	Gap	CPU	Gap
0.5	1	0.2	0	0.3	0	0.4	0	0.6	0	0.6	0	0.7	0
	3	2.9	0	1.3	0	15.4	0	3.4	0	10.0	0	4.1	0
	5	52.8	0	8.8	0	1451.8	0	114.5	0	**	0.13	264.5	0
	7	115.9	0	38.3	0	**	9.88	592.1	0	**	16.42	**	2.68
	9	150.6	0	55.0	0	**	12.22	1460.6	0	**	25.06	**	7.96
	10	93.8	0	43.2	0	**	13.35	1908.3	0	**	27.61	**	9.91
1	1	0.4	0	0.2	0	0.9	0	0.7	0	0.7	0	0.7	0
	3	3.5	0	1.2	0	20.9	0	6.8	0	29.1	0	7.1	0
	5	17.6	0	2.7	0	371.8	0	86.3	0	3506.1	0	734.4	0
	7	19.7	0	3.8	0	1833.9	0	227.3	0	**	9.70	2683.7	0
	9	14.0	0	3.1	0	2696.3	0	272.7	0	**	11.09	**	0.11
	10	13.4	0	3.2	0	2351.9	0	242.9	0	**	10.21	2940.7	0
2	1	0.2	0	0.2	0	0.8	0	0.5	0	1.0	0	0.5	0
	3	2.3	0	1.8	0	8.7	0	4.6	0	7.7	0	8.7	0
	5	1.7	0	1.7	0	16.5	0	23.1	0	64.2	0	77.5	0
	7	2.0	0	1.7	0	197.7	0	33.1	0	1467.0	0	186.8	0
	9	1.4	0	1.0	0	312.7	0	69.5	0	2151.3	0	151.3	0
	10	1.9	0	0.8	0	315.7	0	119.4	0	1808.3	0	146.6	0

MILP and MILP + MC. However, when we slightly increase the problem size to 80 nodes (add 10 customer zones, 5 pos-stations and 5 lockers), MILP fails to solve the model within one hour in some problem instances. The computational time for MILP + MC also grows dramatically. In the 100-node network, when $\alpha = 0.5$ and $\alpha = 1$, both formulations are not efficiently in solving these instances. Therefore, MILP approaches are not suitable for large-scale problems.

Table 3
Performance of MILP + MC and QT-LA. N denotes the number of nodes. C and C^q denote the service level (%) by MILP and QT-LA. CPU denotes the time in seconds used to solve the problem. ** indicates that the instance cannot be solved within 3600s by MILP + MC.

N	α	P	MILP + MC		QT-LA			
			C	CPU	C^q	γ	CPU	Δ(%)
100	0.5	5	31.85	264.5	31.85	1	0.3	0
		7	33.07	**	33.07	1	0.3	0
		10	34.58	**	34.58	1	0.4	0
		15	36.66	**	36.76	0.8	0.6	−0.27
	1	5	52.24	734.4	52.24	0.9	0.4	0
		7	53.70	2683.7	53.70	0.6	0.4	0
		10	55.42	2940.7	55.42	1	0.5	0
		15	58.14	780.0	58.14	1	0.4	0
	2	5	71.69	77.50	71.69	0.6	0.5	0
		7	72.86	186.8	72.86	0.9	0.5	0
		10	73.70	146.6	73.70	0.8	0.7	0.07
		15	74.67	171.8	74.55	0.5	0.8	0.16
400	0.5	10	30.69	**	31.84	1	1.4	−3.61
		20	29.54	**	33.06	1	2.3	−10.65
		30	29.77	**	33.78	1	1.8	−11.87
		40	31.39	**	34.19	1	5.3	−8.19
	1	10	54.17	**	54.24	1	2.0	−0.13
		20	51.17	**	55.93	1	2.9	−8.51
		30	53.93	**	56.79	1	4.7	−4.98
		40	52.96	**	57.37	0.9	14.8	−7.69
	2	10	76.61	**	78.99	1	5.6	−3.01
		20	76.77	**	80.43	0.7	16.3	−4.55
		30	77.85	**	81.24	0.9	32.9	−4.17
		40	77.85	**	81.76	1	58.7	−4.78

5.1.2. Performance of QT-LA

We test the performance of our proposed QT-LA. We use two artificial generated networks, namely, the 100-node (small) network and the 400-node (large) network. The number of customer zones, POP-stations and candidate lockers is 50, 25 and 25 in the 100-node network; and 200, 100 and 100 in the 400-node network.

We first compare the performance of QT-LA with MILP + MC. We set the maximal computational time for MILP + MC to 3600s. Table 3 reports the computational results of 24 problem instances. For QT-LA, γ is selected from {0.5, 0.6, 0.7, 0.8, 0.9, 1}. For all instances, we start QT-LA with two initial solutions: (1) x^0 and r^0 are fixed at $x_j^0 = 1, \forall j \in J$, and $r_k^0 = 1, \forall k \in K$; (2) x^0 and r^0 are fixed at $x_j^0 = 0, \forall j \in J$, and $r_k^0 = 1, \forall k \in K$. Δ denotes the relative difference between the objective by MILP + MC and the objective by the best QT-LA. It is computed by $(C - C^q)/C^q$. A positive value of Δ indicates MILP + MC finds a better solution than QT-LA; vice versa.

For small-scale problems, QT-LA is much faster and yields high-quality solutions. In the 100-node network, MILP + MC is able to solve most of the instances to global optimality. In this case, the maximum Δ is only 0.16%, meaning that QT-LA finds near-optimal solutions with minimal computational efforts (CPU is less than 0.8s). For large-scale problems, QT-LA is a more reliable method that requires significantly less computation load and leads to better solution quality as well. In the 400-node network, Δ is negative in all instances. In the worst case, Δ drops to − 11.87%. The computational time by QT-LA is still not significant.

To conclude, QT-LA is an efficient algorithm that is suitable for large-scale problems.

5.2. Case study

To illustrate the usefulness of our model, we conduct a case study based on the POP-station network and demand zones of Singapore.

5.2.1. Data collection and the impact of P

Four data sets are described below.

- Demand and demand zone. We assume the demand comes from the residential area. Demand nodes were formed based on the sub-clusters demarcated by Singapore's Urban Redevelopment Authority (URA). The size of each cluster and the number of residents are retrieved from a publication by the Department of Statistics (DOS) Singapore.[2] There are 272 residential clusters. The demand is assumed proportional to the number of residents.
- POP-stations. 137 POP-stations located around Singapore are identified. The address of each POP-station is obtained from Singapore Post's official webpage.
- Potential locker facilities. 139 shopping centers are chosen as potential locations. The list of shopping centers is extracted from Shopping.sg, a shopping guide that provides extensive information on shopping centers in Singapore. Using Google maps, the addresses of these shopping centers are retrieved and added to the list.
- Distance matrix L. We use the Google's API URL to calculate accurate walking distances between nodes. The distance is measured in kilometers with 3 decimal.

That is, $|I| = 272$, $|J| = 139$ and $|K| = 137$. The deterministic part of utility is defined as $v_{im} = -\alpha L_{im}, \forall i \in I, m \in J \cup K$. For the value of $\tilde{a}$, $\tilde{a}_{im} = 1$, if $L_{im} \leqslant 1$;$\tilde{a}_{im} = 0.5$, if $1 < L_{im} \leqslant 1.5$;$\tilde{a}_{im} = 0.2$, if $1.5 < L_{im} \leqslant 2$; and $\tilde{a}_{im} = 0$, if $L_{im} > 2$.

We first consider the problem where P POP-stations shall be closed and P locker facilities are to be located, namely, (8) and (9) are equality constraints. We test the model under different P values and α values. Note that QT-LA is a heuristic algorithm whose convergent point may be affected by the initial condition. To obtain a service level as high as possible, we implement the multi-started QT-LA. For each problem instance, QT-LA is started from 10 different initial solutions. After the algorithm terminates, we select the best solution from the pool.

Fig. 2 depicts the results. The service level C first increases with P to a certain maximum and then decreases. The maximum improvement in the service level occurs when P is near 60 for all values of α. This implies that some candidate lockers are located in the sites closer to the customer zones than the POP-stations. Opening these lockers instead of some remote POP-stations will improve the overall service level. In general, the service level hardly improves when P increases from 25 to 60. For examples, under $\alpha = 4$, C values for $P = 25$ and $P = 60$ are 77.19% versus 77.56%. Under $\alpha = 10$, C values for $P = 25$ and $P = 60$ are 79.43% versus 79.49%. These results suggest that replacing 60 existing POP-stations with 60 carefully selected locker facilities will not lead to an acceptable increase in the service level compared to only replacing 25 POP-stations. In practice, the latter option is advantageous because the effort for making the change is much smaller. On the other hand, customers may be accustomed to the current network and facilities. Closing a large number of POP-stations may result in their discomfort because their frequently visited POP-stations may be closed and they have to adjust the change. Therefore, it is reasonable to replace only a moderate number of POP-stations. From Fig. 2, an appropriate P value can be selected from 20 to 30.

[2] https://www.singstat.gov.sg/-/media/files/publications/population/population2016.pdf.

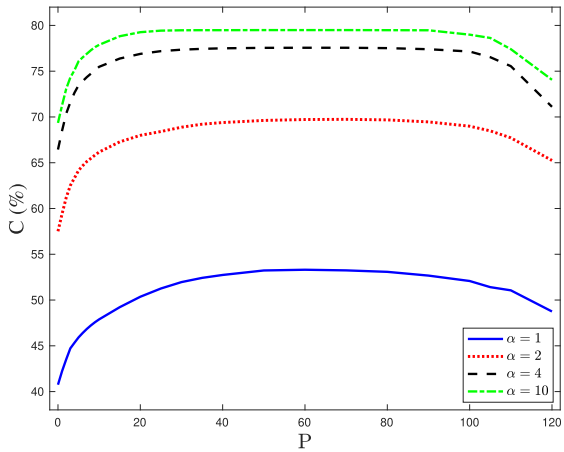

Fig. 2. Case study. Service level C versus P.

Table 4
The set of closed POP-stations suggested by the solution, namely, $\widetilde{K}_\alpha$.

P	$\alpha = 1$	$\alpha = 2$	$\alpha = 4$	$\alpha = 10$
1	{62}	{88}	{136}	{132}
3	{15, 33, 88}	{2, 88, 136}	{41, 93, 136}	{93, 132, 135}
5	{15, 33, 82, 88, 129}	{2, 9, 36, 88, 136}	{9, 41, 93, 131, 136}	{0, 9, 120, 135, 136}

5.2.2. Impact of α

Next, we discuss the impact of α on the service level and the solution. From Fig. 2, when α increases, C will increase. Intuitively, a large value of α implies that the customers show high preference to their nearest facility, meaning that only a small proportion of customers will visit the non-nearest facilities. The service level provided by the alliance will thus be high.

Let $\widetilde{K}_\alpha = \{k | r_k = 0, \forall\, k \in K\}$, that is, $\widetilde{K}_\alpha$ is the set of closed POP-stations suggested by the model. Table 4 shows the sets corresponding to 3 different values of P. We see that the solutions are different when α values are different. In particular, for all values of P, $\widetilde{K}_1 \cap \widetilde{K}_{10} = \varnothing$, indicating that $\widetilde{K}_1$ is completely different from $\widetilde{K}_{10}$. Therefore, in order to make the best decision in practice, the alliance needs to carefully estimate the parameter α.

In our final experiment, we consider the scenario where the alliance wants to close 20 POP-stations and open an unrestricted number of locker facilities to maximize the service level, that is, we set $P = 20$ and remove constraint (8). As shown in Table 5, as α increases, the model suggests to open more lockers and the total number of facilities increases. This is reasonable because the customers' choices are "random" in the case of a small α value. Given too many facilities, customers will "randomly" select one to visit. In many cases, they may walk a long distance, resulting in a lower service level. Conversely, with a large α value, customers will almost always patronize the nearest facility. It is then reasonable to operate more facilities (than the case with a small α value) to gain proximity to customer zones.

According to Table 5, the model does not recommend to open $|J|$ lockers. This result is fundamentally different from the traditional coverage model where the solution is to open $|J|$ lockers if the maximum number of lockers (that are allowed to open) is unrestricted and the cost is not considered. The reason for the difference lies in the model assumptions. More specifically, the traditional coverage model assumes that the operator determines the customer-facility allocation or that customers always select the nearest facility. However, this assumption is not valid in e-commerce because customers take the initiative. Meanwhile, the choice behaviors are typically probabilistic rather than deterministic (Gul et al., 2014; Haase and Müller, 2014; Ljubić and Moreno, 2018). Therefore, the MNL based on random utility theory is applied to forecast the choice. Under MNL, if we open an excess number of facilities, the chance of a customer patranizing the nearest one will decreases. In some cases, we could observe an interesting phenomenon, that is, the alliance intends to enhance the service by opening new facilities, but the result is the opposite.

Table 5
Results under $P = 20$ when constraint (8) is removed. C denotes the service level (%); X denotes the number of lockers; and F denotes the total number of facilities.

	$\alpha = 1$	$\alpha = 2$	$\alpha = 4$	$\alpha = 10$
C	50.54	68.07	77.18	79.48
X	29	40	44	51
F	146	157	161	168

To summarize, under different α values, the final decisions could be completely different. When determining where to locate the locker facilities and which POP-stations to close, it is important to consider the customers' choices. The value of α, therefore, should be carefully determined or estimated in advance (e.g. conducting a parameter estimation using survey data). Otherwise, we may operate facilities that ultimately reduce the overall service level.

6. Conclusion

This paper focused on the last-mile delivery problem. We considered a POP-Locker Alliance who operates a set of POP-stations and wishes to improve the service level by opening new locker facilities. We studied the optimal location of locker facilities, considering the customer' preference, with the objective to maximize the overall service level. The customer's choice was modeled by a MNL model. We formulated the problem as a MLFP. To solve the model, we provided two approaches. The first one was the MILP approach. We reformulated our original problem as an equivalent MILP formulation and further strengthened the formulation using McCormick inequalities (MC). Our numerical test suggested that adding MC can significantly reduce the computational time. However, the strengthened MILP cannot scale well on the problem size. To solve large-scale problems, we applied the QT-LA algorithm. In our numerical tests, QT-LA outperformed MILP approach in both solution time and solution quality. Finally, we conducted a case study to illustrate the usefulness of our model. The results highlighted the importance of considering the customers' choice behaviors and the difference between traditional coverage model and our model.

There are limitations. We did not consider the facility capacity. Indeed, capacity is an important factor that can affect the customer's choice behavior. For example, if a locker has no remaining capacity, customers may select other facilities. We assumed that customers are homogeneous in their observable characteristics. This strong assumption can be partially relaxed by dividing customers into different classes according to their characteristics such as age, income and car-ownership. Besides, we measured the service level by a function of distance and the customer's choices were developed based on distance as well. In practice, other factors, such as transportation convenience and proximity to working places, are also important. Future research can consider them in the choice model and propose a more general measurement for the service level. Finally, for large-scale problems, we applied QT-LA. Even though it works well in practice, it is not an exact solution approach. Future studies can develop an exact and efficient solution approach.

CRediT authorship contribution statement

Yun Hui Lin: Methodology, Software, Writing - original draft. **Yuan Wang:** Conceptualization, Methodology, Writing - review & editing. **Dongdong He:** Investigation, Validation. **Loo Hay Lee:** Supervision.

Appendix A

A.1. Proof of Proposition 2

Proof. The upper bound on z_i is equivalent to the lower bound on $\sum_{k\in K}\widetilde{\theta}_{ik}r_k+\sum_{j\in J}\theta_{ij}x_j$. For $i\in I$, we define the auxiliary problem:

$$\begin{aligned} h_i = &\min \sum_{k\in K}\widetilde{\theta}_{ik}r_k+\sum_{j\in J}\theta_{ij}x_j \\ &st.(8)-(11) \end{aligned}$$

By (8), h_i is attained only when $x_j=0,\ \forall\, j$. By (9), we need to select at least $|K|-P$ elements from $\widetilde{\theta}_{in},\ \forall\, n\in K$. To attain h_i, we select the smallest $|K|-P$ elements. □

A.2. Proof of Proposition 3

Proof. For $\forall\, i\in I, j\in J$, define the following auxiliary problem:

$$\begin{aligned} h_{i|x_j=\xi} = &\text{optimize} \sum_{k\in K}\widetilde{\theta}_{ik}r_k+\sum_{j\in J}\theta_{ij}x_j \\ &st.(8)-(11) \\ &x_j=\xi \end{aligned}$$

For $z^u_{i|x_j=1}$, we set "optimize" to "min" and $\xi=1$. The optimal value of this problem is attained only when $x_n=0,\ \forall\, n\in J\backslash\{j\}$. By (9), we need to select at least $|K|-P$ elements from $\widetilde{\theta}_{in},\ \forall\, n\in K$. To attain h_i, we select the smallest $|K|-P$ elements.

For $z^l_{i|x_j=0}$, we set "optimize" to "max" and $\xi=0$. The optimal value is attained only when $r_k=1,\ \forall\, k\in K$. By (8) and $x_j=0$, we need to select at most P elements from $\theta_{im},\ \forall\, m\in J\backslash\{j\}$, to maximize the objective function, i.e., we should select the largest P elements.

Similarly, for $z^l_{i|x_j=1}$, we set "optimize" to "max" and $\xi=1$. The optimal value is attained only when $r_k=1,\ \forall\, k\in K$. By (8) and $x_j=1$, we need to select the largest $P-1$ elements from $\theta_{im},\ \forall\, m\in J\backslash\{j\}$.

A.3. Proof of Proposition 4

Proof. For $\forall\ i \in I,\ k \in K$, define the following auxiliary problem:

$$\begin{aligned} h_{i|r_k=\xi} = & \text{optimize} \sum_{k\in K} \widetilde{\theta}_{ik} r_k + \sum_{j\in J} \theta_{ij} x_j \\ & st.(8)-(11) \\ & r_k = \xi \end{aligned}$$

For $z^u_{i|r_k=1}$, we set "optimize" to "min" and $\xi = 1$. The optimal value of this problem is attained only when $x_n = 0,\ \forall\ n \in J$. By (9) and $r_k = 1$, we need to select at least $|K| - P - 1$ elements from $\widetilde{\theta}_{in},\ \forall\ n \in K\backslash\{k\}$. To attain h_i, we select the smallest $|K| - P - 1$ elements.

For $z^l_{i|r_k=0}$, we set "optimize" to "max" and $\xi = 0$. The optimal value is attained only when $r_k = 1,\ \forall\ k \in K\backslash\{k\}$. By (8), we need to select at most P elements from $\theta_{im},\ \forall\ m \in J$, i.e., to select the largest P elements.

Similarly, for $z^l_{i|r_k=1}$, we set "optimize" to "max" and $\xi = 1$. The optimal value is attained only when we set $r_k = 1,\ \forall\ k \in K$, and select the largest P elements from $\theta_{im},\ \forall\ m \in J$. □

Appendix B

We briefly discuss the impact of γ on QT-LA algorithm. The test problem instance is a 140-node network with $\alpha = 1$ and $P = 30$. In terms of the objective function value, the optimal γ^* is 0.9. QT-LA algorithm is run under 3 values of γ, namely, 0.5 (low), 0.9 (optimal), and 1.0(high). Fig. B.3 shows the convergence of QT-LA algorithm.

For a smaller γ, the algorithm proceeds more smoothly. Intuitively, given a small step size, the changes in parameters will not be dramatical in two consecutive iterations. This also implies the algorithm may need more iterations to converge. From Fig. B.3, QT-LA converges in iteration 8 when $\gamma = 1$, in iteration 8 when $\gamma = 0.9$ and in iteration 18 when $\gamma = 0.5$.

Meanwhile, under different γ values, QT-LA may converge to different solutions. It is thus important to tune the value of γ. One method is to try different values and select the one that returns the best objective. From our numerical studies, this simple idea works well.

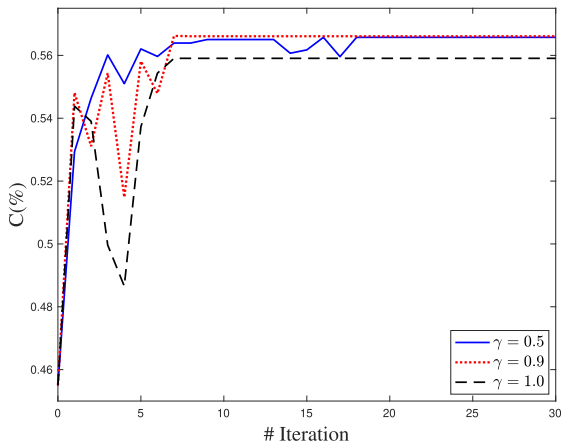

Fig. B.3. Convergence of QT-LA under different γ values. # Iteration denotes the number of iterations.

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